Paper2Agent: Reimagining research papers as interactive and reliable AI agents

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Nature · 2026 - https://doi.org/10.1038/s41586-026-11044-y

Why papers remain difficult to reuse

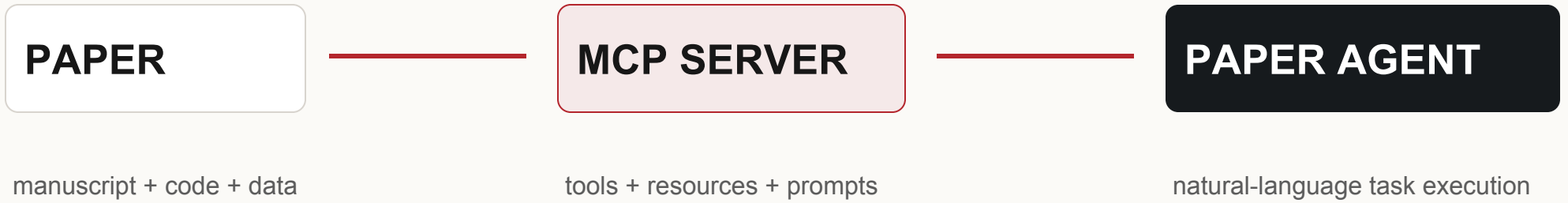
A published method still leaves the reader with a translation problem

Find	Locate the correct repository, data and supplements
Configure	Resolve environments, dependencies, credentials and hardware
Interpret	Map prose and tutorials to valid inputs, parameters and outputs
Adapt	Apply the method to new data without silently changing its meaning

The paper calls this a gap between dissemination and practical use.

CORE IDEA

The paper becomes an executable interface



Beyond retrieval

The agent can call tested functions, reproduce analyses and apply workflows to new inputs.

Paper2Agent construction workflow

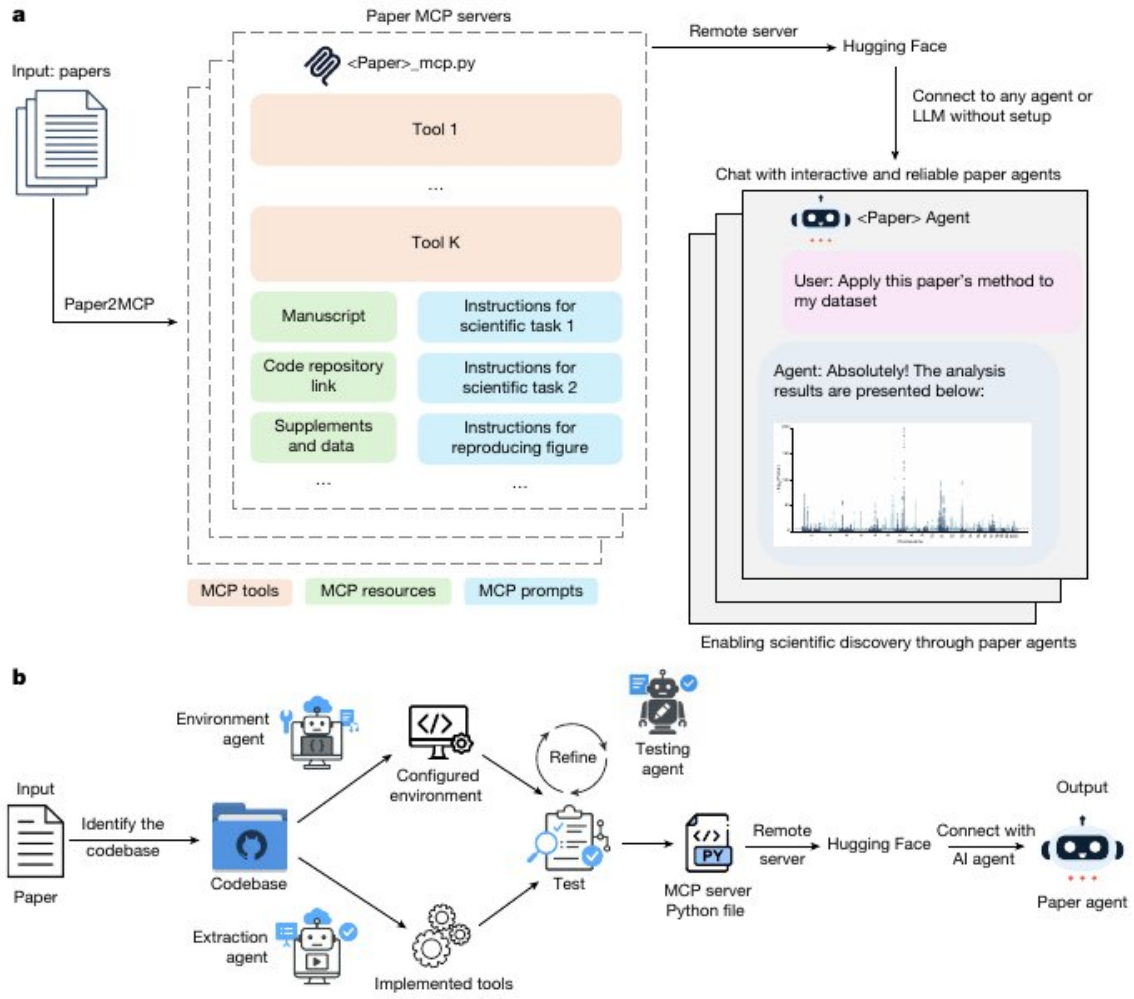


Figure 1: Overview of Paper2Agent. a, Paper2Agent turns research papers into interactive AI agents by building remote MCP servers with tools, resources and prompts. Connecting an AI agent to the server creates a paper-specific agent for diverse tasks. b, Workflow of Paper2Agent. It starts with codebase extraction.

THREE COMPONENTS

What enters the MCP server

TOOLS

Executable functions that wrap the paper's methods

Example: score a genetic variant and save predictions

RESOURCES

Static assets in standard formats

Manuscript, code, supplements, tables, figures and datasets

PROMPTS

Instructions for multi-step scientific workflows

Example: Scanpy preprocessing and clustering order

Traceability: generated tools include references to the original source code.

Validation converts generated code into tested tools

1. **Run tutorials**
   Capture reference files, values and figures

2. **Extract functions**
   Replace hard-coded examples with reusable parameters

3. **Test outputs**
   Check files, numerical tolerance and visual similarity

4. **Repair or exclude**
   Allow up to six attempts, then remove failing MCP tools

Numerical tolerance

3%

floating-point values

Figure threshold

<20

perceptual-hash Hamming distance

Miao et al., Nature (2026), doi:10.1038/s41586-026-11044-y

EVIDENCE

AlphaGenome case study

22

MCP tools generated and validated

~45 min

construction time on a personal laptop

$14

reported one-time construction cost

96.7%

human evaluator agreement

Benchmark accuracy across five runs

Task	Paper2Agent	Claude + Repo	Biomni
Tutorial-derived	**98.7%**	82.7%	37.3%
Novel	**100%**	78.7%	56%
Open-ended	**82.7%**	56.7%	72.2%

The benchmark tests execution plus answer quality. It does not isolate every source of gain.

Source: Miao et al., Fig. 2 and main text

Evidence beyond one repository

74 / 100

computational biology papers successfully agentified

593 / 599

proposed tools passed automated validation

91.2%

accuracy on 300 tutorial-derived questions

Against direct repository access

Paper2Agent: 91.2%

Claude + Repo, Sonnet 4: 80.3%

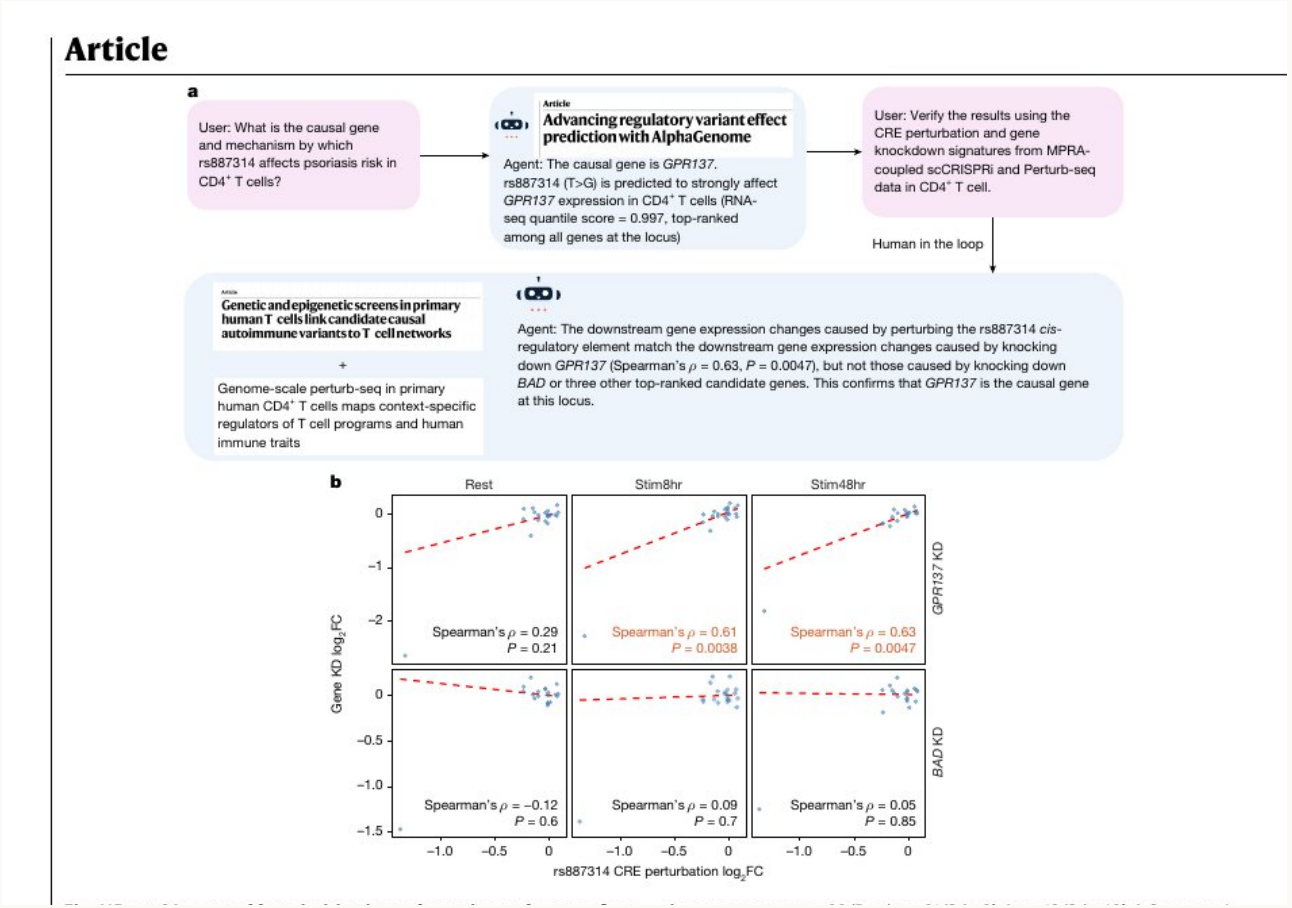
Claude + Repo, Sonnet 4.6: 86.3%

Failure modes

Missing executable code
Missing data or model artefacts
Dependency and environment failures
Scripts that did not generalize

Miao et al., Nature (2026), doi:10.1038/s41586-026-11044-y

Paper agents collaborate across studies



Three agents

AlphaGenome prediction
MPRA-coupled scCRISPRi
CD4+ T-cell Perturb-seq

Result

GPR137 emerged as the probable causal gene for the psoriasis locus rs887314, supported by significant concordance under stimulated conditions.

Human choice remained in the loop: the researcher selected the validation strategy.

Source: Miao et al., Fig. 4 and main text

Strengths of the study

1

Execution grounded in source code

The system calls validated tools instead of generating analysis code from scratch for every query.

2

Explicit failure handling

Repeatedly failing functions are excluded, and unsuccessful papers remain visible in the scale evaluation.

3

System-level comparisons

Benchmarks compare against direct repository access and a scientific agent baseline.

4

Multiple evidence levels

The paper includes reproduction, novel queries, scale testing and a cross-paper discovery example.

Limitations and open questions

Generality

Twenty-six percent of computational biology papers could not be agentified.

Validity

Tutorial agreement tests software fidelity, not whether a method is appropriate for a new dataset.

Benchmark dependence

Accuracy may depend on the selected model, prompts, rubrics and reference answers.

Security and governance

Executing third-party research code and handling private data require stronger operational controls.

Maintenance

Repositories, dependencies, APIs and hosted MCP endpoints will drift after publication.

Scientific authority

Open-ended hypotheses remain human-in-the-loop and should not be treated as validated conclusions.

Inputs required before conversion

1	**Paper**	PDF, URL or local manuscript files
2	**Code**	Official GitHub repository or local source directory
3	**Reference workflow**	A tutorial, notebook, example script or named scientific task
4	**Required assets**	Example data, model weights, API keys and supplementary files
5	**Execution resources**	Python or R, native tools, sufficient memory and any required GPU

Source: Paper2Agent GitHub README, accessed 18 Sep 2026

Paper agent creation

1

Install the skill

Install the complete skills/paper2agent folder in a compatible coding agent.

2

Submit the sources

Provide the paper, repository, associated files and an output directory.

3

Define the scope

Name a tutorial or list the scientific tasks that should become tools.

4

Run conversion

The agents configure the environment, execute tutorials, extract functions and build tests.

5

Review validation

Inspect passing tools, failed or excluded functions, source references and test outputs.

6

Connect the server

Follow USAGE.md to add the local MCP server to Codex, Claude Code or another client.

Run the official example first. Use new data only after the reference workflow reproduces correctly.

Source: Paper2Agent GitHub README, accessed 18 Sep 2026

Example conversion request

PROMPT

Use the paper2agent skill to agentify this paper and its associated files, alongside its code repository.

Paper and associated files: <PAPER_URL_OR_LOCAL_FILES>
Code repository: <GITHUB_URL_OR_LOCAL_PATH>
Output directory: <PROJECT_DIR>

Focus on <TASK, TUTORIAL TITLE OR TUTORIAL URL>.
Report any tool that cannot reproduce the reference output.

After conversion: “Connect the generated MCP server to my coding-agent client using its USAGE.md.”

Adapted from the Paper2Agent GitHub quick start

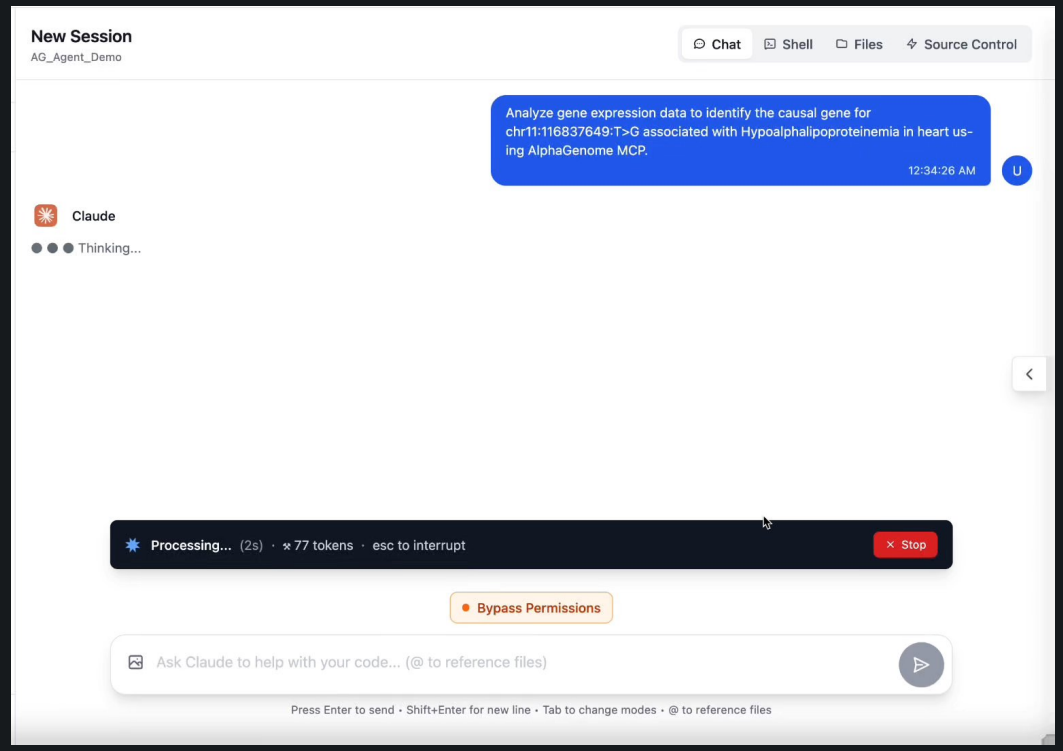
Demo 1 · AlphaGenome

SCIENTIFIC TASK

Interpret a genetic variant associated with hypoalphalipoproteinemia

Example query

Analyze heart gene-expression data with AlphaGenome MCP to identify the causal gene for the variant chr11:116837649:T>G.



What the demo shows

The agent selects AlphaGenome tools, requests tissue-specific predictions, analyzes regulatory effects and returns a biological interpretation without the user writing API code.

Demo video: AlphaGenome_chatbot.mov in the Paper2Agent repository

Demos 2 and 3 · TISSUE and Scanpy

TISSUE

Spatial transcriptomics with uncertainty

Query: Calculate the 95% prediction interval for spatial expression of Acta2.

Inputs: Spatial_count.txt
Locations.txt
scRNA_count.txt

The agent loads the three data sources, predicts spatial expression and reports uncertainty.

SCANPY

Single-cell preprocessing and clustering

Query: Preprocess and cluster the single-cell dataset pbmc_all.h5ad.

Workflow: quality control, filtering, normalization, dimensionality reduction, neighbour graph, Leiden clustering and UMAP.

The MCP prompt preserves the correct order of the analysis steps.

Repository video: TISSUE_chatbot.mov. The README presents the Scanpy query and hosted MCP server.

Source: Paper2Agent GitHub demo section

Takeaways and discussion

1. **Paper2Agent packages code, data, prompts and tests behind a natural-language interface.**

2. **Its strongest contribution is the validation pipeline, not the chat interface alone.**

3. **Agentifiability may become a useful stress test for computational reproducibility.**